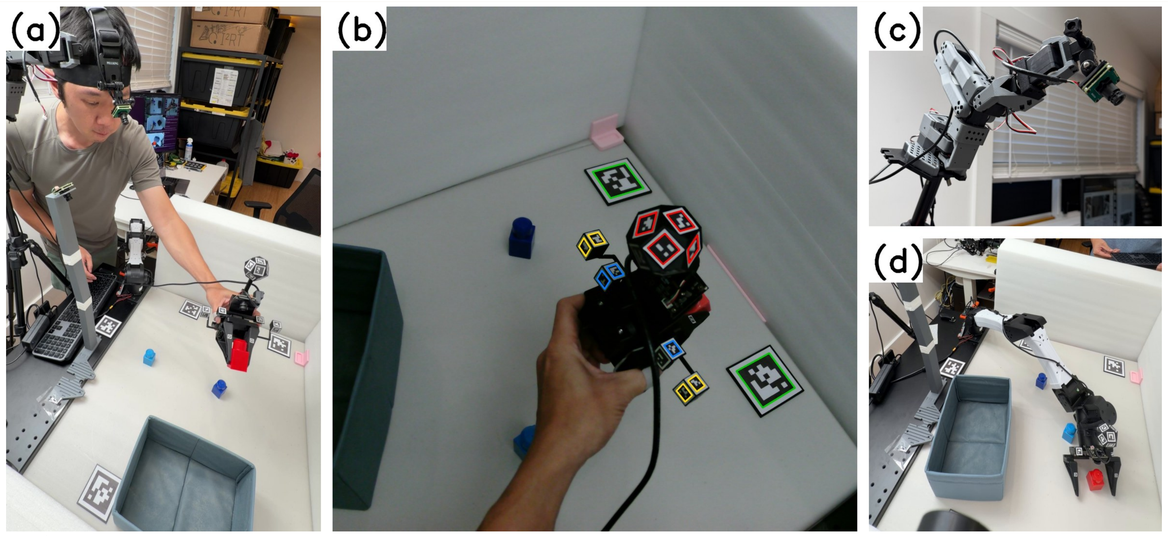
A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

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arXiv:2610.27622
The UMI Arena, CoRL 2026

One head camera: the policy's observation and the pose labels



(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

Why a head cam

- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

Labels from one frame

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

6.5 mm tool point · **1.6°** orientation · **3.5 %** jaw width, of stroke

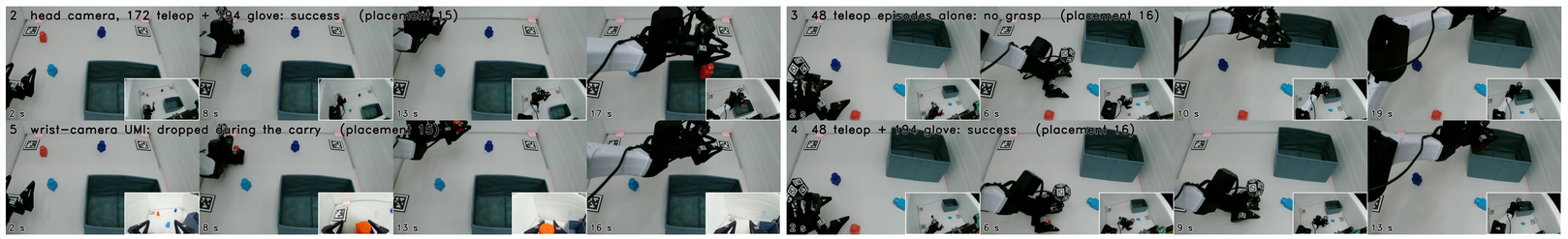
Median error against the robot's encoders; a label on 93–100 % of frames.

The robot's head

- A $122 six-joint arm carries the same camera beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts.

46 mm head pose vs the collector · **5.4°** static camera: 87 mm, 7.0°

Four rollouts



Fixed workspace camera, each policy's own view inset. Left, placement 15: the head-camera policy carries the block to the bin; the wrist-camera policy, seeing only block and wall, drops it mid-carry. Right, placement 16: 48 teleoperated episodes alone never grasp; 48 plus 194 glove demos succeed.

Limits and next

One task, one collector, one tile layout. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.

Collection

$144 glove, camera, strap · **103 g** on the head · **2–3×** demos/min vs teleop · **198** demos in four sessions

Eight policies, one recipe

ACT, joint-space actions, same 194 glove demos in every mix. 25 placements, one trial each.

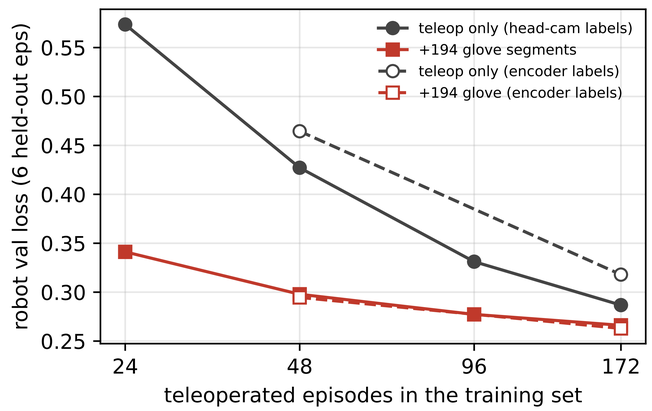
#	Policy (glove labels → observation)	Val	Success
1	teleop only, 172 ep. → head cam	0.318	18/25
2	**device**: head → head, 172 + 194	**0.263**	**20/25**
3	teleop only, 48 ep. → head cam	0.464	10/25
4	**device, low budget**, 48 + 194	0.295	**17/25**
5	wrist UMI: fixed → wrist fisheye	0.257	15/25
6	head-cam labels for encoders	0.328	15/25
7	fixed camera as the system	0.256	12/25
8	head labels → wrist fisheye	0.252	14/25

The view decides how a policy fails

20/25 head camera · **15/25** wrist fisheye (UMI) · **12/25** fixed camera

- Wrist policies **drop the block mid-carry** (12 of 21 failures), always while looking away from the bin. Head policies: 1 of 45.
- Same labels, other view (arms 8 vs 2): 20/25 falls to 14/25. Same view, other labels (8 vs 5): no difference.
- **Head-camera labels** cost 3 % of validation loss against encoders, nothing on the robot (6 vs 1).

194 glove demos ≈ 65–100 teleoperated episodes



Robot validation loss against teleoperation budget, with (dashed) and without the glove set. Added to 48 teleoperated episodes: 10/25 to 17/25, level with 172 alone.